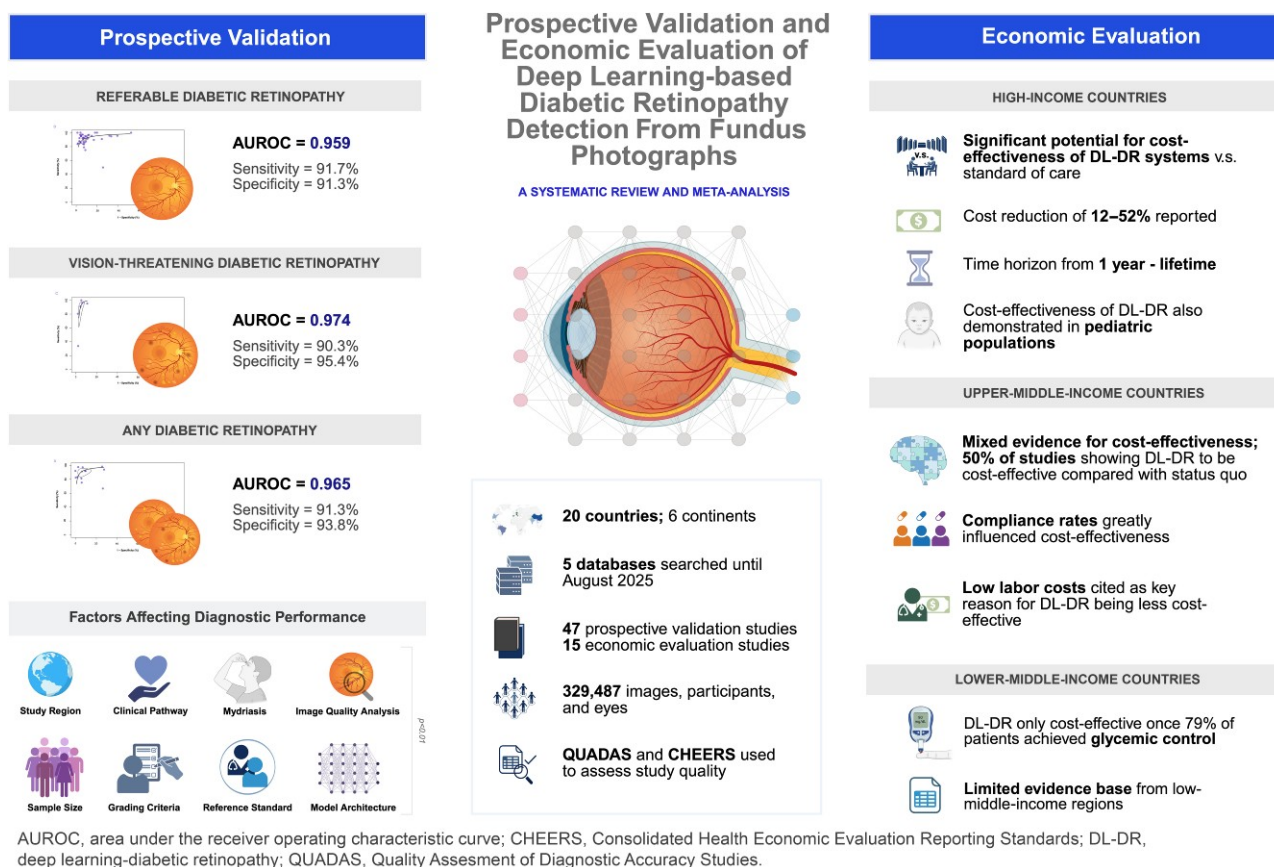


Real-World Prospective Validation and Economic Evaluation of Deep Learning–Based Diabetic Retinopathy Detection From Fundus Photographs: A Systematic Review and Meta-analysis

An Ran Ran, Jennifer Li Ding, Ziqi Tang, Ching Lam, Truong X. Nguyen, Jiaying Zhou, Shuyi Zhang, Danqi Fang, Dawei Yang, Vincent Ng, Duoru Lin, Haotian Lin, Clement C. Tham, Carmen K.M. Chan, Simon K.H. Szeto, Tien Y. Wong, Sobha Sivaprasad, and Carol Y. Cheung

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ARTICLE HIGHLIGHTS

• Why did we undertake this study?

Deep learning (DL) is a promising solution for diabetic retinopathy (DR) screening from fundus photographs; however, real-world modifiers of its diagnostic performance and cost-effectiveness remain poorly characterized, necessitating prospective evidence synthesis.

• What are the specific questions we wanted to answer?

How do DL-DR systems perform in prospective, real-world settings and what is their cost-effectiveness across income settings?

• What did we find?

Across 47 prospective validation studies, DL-DR systems achieved >90% pooled discriminative accuracy. Performance was influenced by region, clinical pathway, mydriasis, image-quality control, sample size, grading criteria, reference standard, and model architecture. Our economic evaluation found DL-DR generally cost-effective, especially in high-income countries.

• What are the implications of our findings?

DL-DR systems can deliver accurate, cost-effective screening. Further evaluation in low-resource settings will facilitate equitable deployment.



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See accompanying article, p. 381.

BACKGROUND

Deep learning (DL) has shown promise in delivering diagnostic and economic benefits for detecting diabetic retinopathy (DR) from fundus photographs (FPs). However, evidence synthesis of model validation in prospective, real-world settings remains limited.

PURPOSE

To assess the feasibility of implementing DL-DR systems using FPs across different countries by synthesizing prospective validation and economic evidence.

DATA SOURCES

Five databases were searched until 13 August 2025.

STUDY SELECTION

Studies prospectively assessing diagnostic performance and/or studies conducting economic analyses of DL-DR systems using FPs were selected.

DATA EXTRACTION

Characteristics of all studies, performance parameters of prospective validation studies, and economic outcomes of economic analysis studies were extracted.

DATA SYNTHESIS

Forty-seven studies were included in the meta-analysis. The pooled performance was the highest in detecting vision-threatening DR (area under the receiver operating characteristic curve [AUROC] 0.974), followed by any DR (AUROC 0.965), then referable DR (RDR) (AUROC 0.959). Study region, clinical pathway, mydriasis, image quality control, sample size, grading criteria, reference standard, and model architecture significantly affected model performance in RDR detection. Fifteen studies were included in the economic commentary, showing that DL-based DR screening was cost-effective in high-income countries, whereas results in middle-income countries were mixed, depending on compliance rates, glycemic control, and initial costs.

LIMITATIONS

A paucity of studies assessing multiple severities of DR or diabetic macular edema restricted our ability to perform subgroup analyses. Insights into low-income countries were limited by a lack of studies in these regions.

CONCLUSIONS

DL-DR systems using FPs had high discriminative performance in prospective real-world settings and hold promise to improve cost-effectiveness, especially in high-income countries.

Diabetic retinopathy (DR), a complication of diabetes mellitus, remains a leading cause of vision loss globally (1). Importantly, early detection and prompt intervention for the more severe stage of DR—vision-threatening DR (VTDR)—can prevent irreversible visual impairment (2). To address this, national DR screening programs using digital retinal fundus imaging have been implemented in several high-income countries, typically at the primary care level (3,4). However, current programs are often capital and labor intensive, and face ongoing implementation and sustainability challenges. For example, most programs rely on trained personnel to obtain and interpret fundus photographs (FPs). Manpower shortages, interindividual differences in interpretation, and missed cases of retinal pathologies may all limit screening efficacy (5). Thus, there is an unmet need for a more efficient, cost-effective, and robust screening solution that can be applied globally.

Over the past decade, artificial intelligence (AI), particularly deep learning (DL), has emerged as a highly promising technology solution to fill this gap. Although multiple DL-based DR detection systems using FPs have been piloted in clinical settings (6–8), the generalizability and consistency of their diagnostic accuracy in prospective, real-world settings remain uncertain. Prospective, real-world validation is crucial before deployment, because model performance can vary based on contextual factors, including population-level characteristics, health care settings, and resource availability (9). Contextual variability not only exists in systems' diagnostic performance but also in their economic value. Yet, the real-world economic viability of DL-DR systems remains poorly understood because the evaluation of DL-DR systems has traditionally focused solely on diagnostic performance measures, such as sensitivity and specificity. Previous reviews have provided insight into DL

systems for DR detection from FPs, particularly highlighting the importance of head-to-head validation and assessing cost-effectiveness (9). However, there remains a lack of quantitative evidence synthesis demonstrating the discriminative performance of DL-DR systems using FPs in prospective, real-world settings. A comprehensive commentary on the economic impact of DL-DR systems is also warranted to address questions regarding the feasibility of applying DL in screening programs across countries with differing population needs and health care system capacities.

Here, we report on a systematic review and meta-analysis conducted to synthesize current evidence and quantify the pooled discriminative performance of DL-DR systems using FPs in prospective settings. Additionally, we provide an economic commentary and propose directions to address deficiencies in current economic evaluations of DL-DR systems using FPs.

RESEARCH DESIGN AND METHODS

We undertook a structured search across five databases (Cochrane Library, PubMed, Web of Science, Scopus, and Google Scholar) until 13 August 2025. One round of a systematic forward and backward search for references was conducted. Study authors were contacted for unpublished studies or additional data. Full details of the search strategy are described in Supplementary Table 1. This study was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) protocol guidelines (10) and adhered to the principles of the Declaration of Helsinki. The review was registered in PROSPERO (CRD42023344298).

All search results were screened and synthesized using EndNote. After removing duplicates, two reviewers (A.R.R. and J.L.D.) screened all retrieved titles and abstracts independently. Conflicts were resolved by consensus. The two reviewers

then independently assessed the full texts of eligible reports. At any stage, if the two reviewers could not achieve consensus, the senior reviewer (C.Y.C.) evaluated the records and made the final decision.

Eligibility Criteria

Inclusion criteria for the prospective validation studies were 1) prospectively validating the discriminative performance of DL-based algorithms to detect DR, including any DR, referable DR (RDR), and VTDR, from FPs captured by conventional tabletop retinal cameras; and 2) providing outcomes of binary classification tasks or multiple-outcome classification. The definition of RDR was based on the International Classification of Diabetic Retinopathy (ICDR) scale (11) as having moderate nonproliferative DR, or worse. Details of the conversion criteria used to harmonize data from studies using varying grading systems can be found in Supplementary Table 3.

Characteristics of excluded prospective validation studies were 1) screened for eye diseases other than DR; 2) used imaging modalities other than conventional tabletop retinal cameras; 3) solely used non-DL-based methods to evaluate FPs; 4) did not specify that the used algorithm was DL; 5) used retrospectively collected data sets for model validation; 6) did not report or the authors were unable to provide necessary metrics for meta-analysis (i.e., sensitivity, specificity, numbers of total and disease-positive participants, eyes, or images); 7) studies published in non-English languages; 8) studies with non-human subjects; 9) not peer-reviewed original research articles or articles without available full text; and 10) studies that focused predominantly on participants <18 years of age.

For the economic commentary, peer-reviewed articles that reported upon the economic evaluation of DL-DR systems using FPs with either prospective data sets, retrospective data sets, or hypothetical cohorts were eligible. Studies

were classified by the income status of the country in 2023, calculated using the World Bank Atlas method and reported in U.S. dollars (12). Exclusion criteria were 1) studies published in languages other than English or with nonhuman subjects, and 2) non-peer-reviewed original research articles or those without available full text.

Full details of the quality assessment, data extraction process, and income-level definitions are described in Supplementary Material, Part 1.

Data Analysis

We conducted a random effects meta-analysis of prospective validation studies to evaluate the discriminative performance of DL-DR systems across varying disease severities, including any DR, and/or RDR, and/or VTDR. The 2×2 matrix values (true positives, true negatives, false positives, and false negatives) from each study were pooled to calculate the discriminative performance using RStudio (version 1.2.5019). In studies in which the 2×2 matrix values were not directly stated, we used metrics, including sensitivity, specificity, and numbers of disease-positive participants, eyes, or images, to calculate the corresponding 2×2 matrix values in Review Manager 5.4. Further information on data harmonization is listed in Supplementary Table 4.

The area under the receiver operating characteristic curve (AUROC) was calculated using a hierarchical model. We reported pooled discriminative performance using the following metrics: AUROC, sensitivity, specificity, diagnostic odds ratio (DOR), along with 95% confidence intervals (CIs). The DOR is a widely recognized summary measure calculated using the following equation; higher values indicate greater ability to distinguish between diseased and non-diseased individuals (13):

$$\text{DOR} = \frac{\text{True Positive} / \text{False Negative}}{\text{False Positive} / \text{True Negative}}$$

The risk of publication bias was initially assessed from conventional funnel plots by visually inspecting for asymmetry and then further evaluated using Deek's method (14).

Sensitivity analyses were performed to assess the impact of outlier studies. We also undertook subgroup analyses for RDR detection based on eight categories: 1) study region (low- and low-middle-income

countries vs. upper-middle and high-income countries); 2) clinical pathway (primary care vs. secondary and/or tertiary care); 3) mydriasis (no mydriasis vs. partial/full mydriasis); 4) image quality assessment (automated or partially automated vs. manual); 5) sample size (large vs. small, using the median number of participants as the cutoff [$n = 848$]); 6) grading criteria (ICDR vs. non-ICDR); 7) reference standard (ophthalmologists vs. trained graders); and 8) model architecture (multiple vs. single convolutional neural network [CNN]). The statistical significance of AUROC values between subgroups was computed using VassarStats (15).

For the economic evaluation studies, we conducted a systematic review, summarized key measurements based on the requirements of Consolidated Health Economic Evaluation Reporting Standards for Interventions That Use Artificial Intelligence (CHEERS-AI) (16), and provided an economic commentary.

Data and Resource Availability

All data used for the study have been included in the manuscript and supplementary material.

RESULTS

Figure 1 depicts the study selection process as a PRISMA flow diagram. Eventually, 47 studies were included in the meta-analysis component, and 15 studies were included in the economic commentary. The Supplementary Material, Part 2, shows the full list of all included studies, reasons for exclusion (Supplementary Table 2), quality assessment (Supplementary Figs. 1 and 2), and the geographic distribution of the studies included (Supplementary Fig. 3).

Characteristics of Prospective Validation Studies

Basic characteristics of the 47 prospective validation studies are summarized in Table 1. Of the 47 studies, 39 reported on RDR, 12 reported on any DR, 6 reported on VTDR, and 2 reported on diabetic maculopathy. Twenty-three studies originated from Asia, 13 from Europe, 4 each from North America and Australia/Oceania, 2 from Africa, and 1 from South America.

Thirty-four studies included image-quality analysis, of which 19 implemented a DL-assisted quality screening module or equivalent. Camera types varied greatly

depending on the study location; the most common cameras were Topcon NW400 and Canon CR-2. A total of 35 studies reported the imaging field of view, ranging from 40° to 200° , with the most common angle being 45° . In terms of the choice of DL system, commonly used models included the EyeArt system (versions 1.0 and 2.0) and IDx-DR (1.0 and 2.0).

DL-DR Performance in Prospective Validation

Performance Across Varying DR Severities

Supplementary Fig. 4A–C illustrates the forest plots for studies detecting any DR, RDR, and VTDR, respectively. Eligible studies were then further compiled into hierarchical summary ROC (HSROC) plots to determine the pooled discriminative performance. Supplementary Fig. 5A–C depict the HSROC plots for any DR, RDR, and VTDR, respectively.

As shown in Table 2, the pooled AUROC for detecting any DR was high, at 0.965, with a pooled sensitivity of 91.3% (95% CI 85.4–95.0) and a pooled specificity of 93.8% (95% CI 87.2–97.1). The pooled discriminative performance of DL in detecting RDR was slightly lower, with an AUROC of 0.959, and a pooled sensitivity and specificity of 91.7% (95% CI 89.5–93.4) and 91.3% (95% CI 88.6–93.5), respectively. VTDR detection had the highest pooled AUROC (0.974) but possessed a lower and more varied sensitivity of 90.3% (95% CI 69.3–97.5), and specificity of 95.4% (95% CI 92.4–97.2).

In our assessment of publication bias, though some funnel plot asymmetry was observed among studies detecting RDR (Supplementary Fig. 6), the interpretation is limited by heterogeneity across studies included in the meta-analysis ($I^2 = 14.6\%$ for RDR detection). In diagnostic test accuracy meta-analyses, such asymmetry may result from clinical and methodological differences, such as variations in populations, thresholds, or settings, rather than publication bias alone (17). Subgroup analyses were thus conducted to explore potential sources of heterogeneity.

Sensitivity Analysis

Based on the results of the forest plot shown in Supplementary Fig. 4B, we conducted a sensitivity analysis by removing eight studies with either sensitivity or specificity $<80\%$. The results showed that excluding these studies did

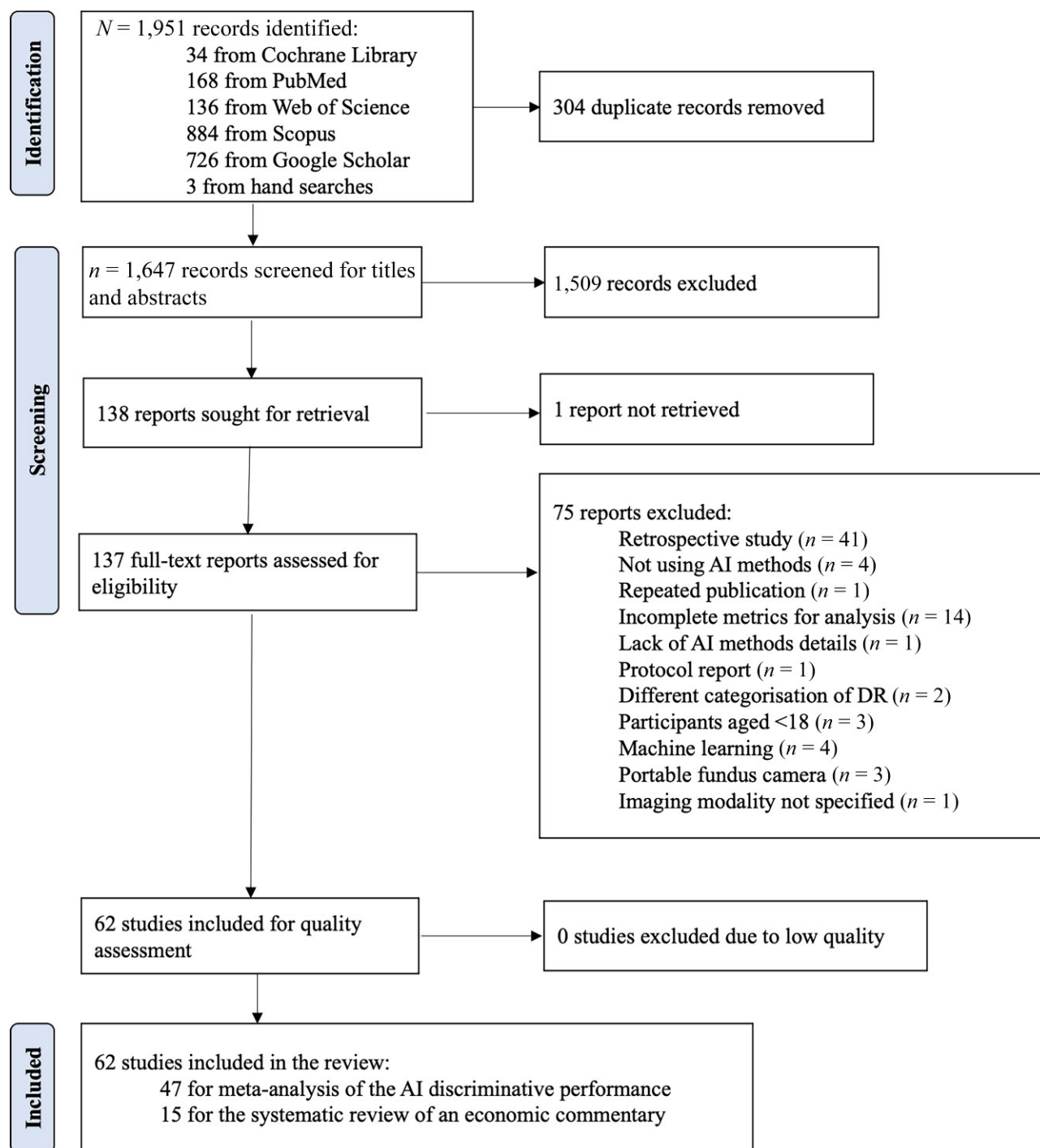


Figure 1—The Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) flow diagram of included studies.

not greatly affect the pooled performance (AUROC 0.965; sensitivity of 91.6% [95% CI 89.4–93.4]; specificity of 93.3% [95% CI 91.3–94.9]). Thus, we continued to include these studies in further subgroup analyses.

Subgroup Analysis for RDR Detection

Due to an insufficient number of studies reporting outcomes of DR and VTDR,

we were only able to perform subgroup analysis for RDR ($n = 39$ studies) (Table 3). Thirty-four studies (87%) were conducted in upper-middle and high-income countries; the rest were conducted in low- and lower-middle-income countries. Studies were conducted in a variety of different clinical settings, with 10 studies (26%) conducted in primary care. The median number of participants, eyes,

and images was 848, which was used as a cutoff point for subgroup analysis.

Comparisons between the AUROC values of subgroup pairs were all statistically significant at $P < 0.01$. Studies based in upper-middle- and high-income countries had significantly higher AUROC (0.957) than those in low- and lower-middle-income countries (0.916). Studies conducted in secondary care (i.e., general

Table 1—Characteristics of included prospective studies on DL-based DR detection

Basic information			Image acquisition			Model development and capabilities			Comparator			Screening setting and workflow		
Reference	Country	Imaging device(s)	Imaging field of view	Mydriasis, yes or no	Architecture	Cross-validation/splitting	Fine tuning	Screening for multiple diseases	Diagnostic criteria	Reference standard	Adjudicated by ≥ 2 graders?	Clinical pathway	Image quality assessment	Time to result
Any DR ($n = 6$ studies)														
Bajwa A, 2022	Pakistan	Not specified	Not specified	Not specified	CNN, not specified	80% training, 20% validation split in phase 1; separate real-time DR data for testing in phase 2	Not specified	No	Not specified	Grading by clinical expert	Not specified	Eye hospital	Automated	Not specified
Cao J, 2022	China	Daytona	200°	Nonmydriatic	ResNetXt-50	60% training, 20% validation, 20% testing, with separate external validation data set	Not specified	Yes	ICDR	Grading by senior ophthalmologist	Not specified	Hospital	Manual	Not specified
Cao S, 2023	China	RetiCam 3100	At least 45°	Nonmydriatic	Transfer learning	Not specified	Not specified	Yes	ICDR	Grading by two ophthalmic chief clinicians with more than 20 years of clinical experience	Yes	Hospital clinic	Manual	Not specified
Hao S, 2022	China	Zeiss VISUCAS500	45°	Nonmydriatic	Inception V3	For lesion detection: 8.8% of images used for testing, 91.2% of images used for training. Within the test set images, 73% were used as the training set for the screening model, and 27% used as the test set for the screening model.	Not specified	Yes	ICDR	Grading by two ophthalmologists with >10 years of experience	Yes	Community hospital	N/A	Not specified
Mbaye S, 2025	Senegal	Topcon TRC-NW8	45°	Mydriatic	CNN, not specified	Not specified	Not specified	No	American Academy of Ophthalmology Grading Standards	Grading by two specialist clinicians	Not specified	Hospital	Not specified	Not specified
Sin M, 2025	Czech Republic	Canon CR-2 AF	Not specified	Nonmydriatic	EfficientNet V2	Not specified	Not specified	No	ETDRS	Grading by three diabetic retinopathy board members	Not specified	Primary diabetes care centers	Automated	Not specified
RDR ($n = 30$ studies)														
Abrahamoff M, 2018	U.S.	NW400, Topcon Medical Systems, Oakland, NJ	45°–60°	Nonmydriatic	IDx-DR with multilayer CNN	Not specified	No; model was locked before validation	No	ETDRS	Three graders from experienced grading center	Not specified	Primary care	Automated	Not specified
Abreu-Gonzalez R, 2025	Spain	Topcon camera (NW200, NW400 or equivalent) and, when available, a ZEISS camera	45°	Nonmydriatic	EfficientNet-B3	80% training, 20% validation	Not specified	No	ICDR	Three retina specialists	Yes	Mixture of primary, secondary, and tertiary care	Manual	Not specified
Bellefleur V, 2019	Zambia	CenterVue	45°	N/A	Two CNN combined: VGGNet architecture and ResNet	Not specified	Not specified	No	ICDR	Two graders with 5 years of experience in diabetic retinopathy grading	Not specified	Community program	Manual	Not specified

Continued on p. 515

Table 1—Continued

Basic information			Image acquisition		Model development and capabilities				Screening for multiple diseases		Comparator		Screening setting and workflow		
Reference	Country	Imaging device(s)	Imaging field of view	Mydriasis, yes or no	Architecture	Cross-validation/ splitting	Fine tuning		Diagnostic criteria	Reference standard	Adjudicated by ≥2 graders?	Clinical pathway	Image quality assessment	Time to result	
Bhuiyan A, 2021 (1)	India	Topcon TRC NW6, CentreVue, Eidon	45°	Nonmydriatic	CNN, not specified	70% training, 30% validation	Not specified	Yes	ICDR	Three expert graders	Not specified	Retina practices, ophthalmic, and primary care clinics	Automated	Not specified	
Bhuiyan A, 2021 (2)	India	CentreVue	45°	Nonmydriatic	Xception, Inception-V3, Resnet-V2	70% training, 12.5% testing, and 17.5% validation	Not specified	No	ICDR	Grading by an ophthalmologist	Not specified	Primary care clinics	Automated	Not specified	
Cicinelli M, 2024	Italy	iCare DRSpus confocal LED camera (CentreVue, Padua, Italy)	45°	Mydriatic	EyeArt v2.1.0	Not specified	Not specified	No	ICDR	Grading during dilated fundus examination by two trained retinal specialists	Yes	Hospital	N/A	Not specified	
Dong L, 2022	China	Canon CR-2 AF, Kowa Nonmyd α-Dili, Topcon TRC-NW200, Topcon TRC-NW300	45°	Nonmydriatic	CNN, not specified	5:1 split of data set into training and testing	Not specified	Yes	ETDRS	Grading by certified retinal specialists with >5 years of clinical experience	Yes	Public medical screening centers and hospitals	Manual	Time to assess each image by: AI: 0.30 s; certified ophthalmologists: 10.88 s; junior retinal specialists: 8.68 s; retinal specialists: 7.68 s	
Dow E, 2023	U.S.	Topcon NW400	45°	Nonmydriatic	IDx-DR	Not specified	Not specified	No	ICDR and ETDRS	In-person eye examination by fellowship trained retina specialist	No	Community primary care clinics and endocrine clinics	Mixed	Not specified	
Gu C, 2023	China	Topcon NW300, RetiCam 3100	45°–50°	Nonmydriatic	Yolo-V3, EfficientNet-B3	5:1 split of training data set; internal validation data set	Not specified	Yes	ICDR	Grading by two retinal specialists with 5–10 years of experience	Not specified	Primary care clinics	Manual	Not specified	
Gulshan V, 2019	India	NM TRC; Topcon Medical Systems; 3methra; Forus Health	40°–45°	Nonmydriatic	Inception-V3 architecture; also used Inception-V4 to improve model during prospective validation	Not specified	Yes	No	ICDR	Grading by one trained grader and one retina specialist	Yes	Tertiary eye care centers	N/A	Not specified	
Heydon P, 2021	U.K.	N/A	N/A	N/A	EyeArt 2.0	Not specified	Not specified	No	NHS	Grading by one primary grader, and for a subset of patients, additional grading by secondary grader, with an arbitration grader to review discrepancies between primary and secondary graders	Not specified	NHS Diabetic Eye Screening Clinics	None	Not specified	
Keel S, 2018	Australia	CentreVue	45°	Nonmydriatic	Four DL models using Inception-V3 architecture	88% of data set for training, 12% for internal cross-validation	Not specified	No	ICDR	Manual grading by retinal grading center	Not specified	Hospital setting	Automated	Mean time for automated screening model: 6.9 min	

Continued on p. 516

Table 1—Continued

Basic information			Image acquisition		Model development and capabilities				Comparator		Screening setting and workflow			
Reference	Country	Imaging device(s)	Imaging field of view	Mydriasis, yes or no	Architecture	Cross-validation/ splitting	Fine tuning	Screening for multiple diseases	Diagnostic criteria	Reference standard	Adjudicated by ≥2 graders?	Clinical pathway	Image quality assessment	Time to result
Li B, 2024	China	Kowa Nonmyd 7, Visucam 224, Canon CR2	N/A	Nonmydriatic	SeResNet50	94.8% training set/ 1.34% validation set/ 3.84% test set	Not specified	Yes	ICDR	Grading by 5 expert ophthalmologists, with a sixth expert ophthalmologist as an arbitrator	Not specified	Tertiary hospital	Mixed	Not specified
Li F, 2022	China	Topcon TRC-NW400, Syeye RetCam 3100, Zeiss	45°–50°	Nonmydriatic	DenseNet121	Not specified	Not specified	Yes	COS, then converted into ICDR	Grading by two certified ophthalmologists with 6 and 10 years of clinical experience	Yes	Hospital	Mixed	AI reporting time: 30 s
Li N, 2022	China	KOWA nonmyd WX	45°	Nonmydriatic	VoxelCloud with two neural networks: DR classification network and network quality control	Not specified	Fine-tuned on domestic data set of 1,184 images	No	ICDR	Grading by one certified retinal specialist with more than 12 years of experience	Not specified	Hospital	Automated	From beginning of retinal fundus image acquisition to output of results: 2 min
Lin DR, 2021	China	Canon, Topcon, Syeye, Crystalvue, Zeiss, Nidek, Kowa	45°–50°	Nonmydriatic	Inception-ResNetV2 on TensorFlow	10:1 split of training: tuning data sets	Not specified	Yes	ICDR	Grading by three ophthalmologists	Not specified	Tertiary hospital	Manual	Not specified
Musetti D, 2024	Italy	SD-OCT, DRI 3D OCT-2000, Topcon	N/A	Mixed; mydriasis was applied if needed	IDx-DR	Not specified	Not specified	No	ICDR	Grading by two retinal specialists	Yes	Hospital	Manual	Not specified
Noriega A, 2021	International/ Mexico	N/A	N/A	Nonmydriatic	16 convolutional layers	Not specified	Not specified	No	ICDR	Grading by three retina specialists	Not specified	Hospital	Manual	Not specified
Platti A, 2024	Italy	iCare DRSpplus	45°	Nonmydriatic	RetCAD	Not specified	Not specified	Yes, but capability of algorithm to detect other diseases not explored	ICDR	Grading by general ophthalmologist	Not specified	Diabetes center and eye center	Mixed	Time taken to process four images: 1 min
Sarao V, 2020	Italy	Topcon TRC-NW8	45°	Nonmydriatic	EyeArt 2.1	Not specified	Not specified	No	NHS	Grading by two independent retina specialists	Yes	Eye care clinic	Manual	Not specified
Scheetz J, 2021	Australia	CenterVue, Canon CR-2 AF, Topcon 3D OCT1 Maestro	45°	Nonmydriatic	Inception v3	Not specified	Not specified	No	NHS	Grading by two retina specialists	Not specified	Outpatient hospital clinics, and service clinics	Automated	Not specified
Sedova A, 2022	Austria	Optomap	45°	Nonmydriatic	IDx-DR	Not specified	Not specified	No	Modified ETDRS, converted into ICDR	Grading by two NHS-certified retinal graders	Not specified	Controlled clinical setting (pilot study) hospital clinics	Automated	Not specified
Skavas C, 2024 (1)	Germany	DRS Plus, Icare Finland Oy, Finland	45°	N/A	Efficient-Net	Not specified	Not specified	Yes	ICDR	Human grader	Not specified	Outpatient hospital clinics	Automated	Time taken for AI to grade image: 30 s
Skavas C, 2022 (2)	Germany	Topcon TRC-NW400	45°	Nonmydriatic	RetCAD software v1.3.1	Not specified	Not specified	Yes	ETDRS	Grading by one expert reference examiner; second examiner used in disagreement between AI and reference examiner	Not specified	Outpatient hospital clinics	Automated	Not specified

Continued on p. 517

Table 1—Continued

Basic information			Image acquisition		Model development and capabilities				Comparator			Screening setting and workflow		
Reference	Country	Imaging device(s)	Imaging field of view	Mydriasis, yes or no	Architecture	Cross-validation/ splitting	Fine tuning	Screening for multiple diseases	Diagnostic criteria	Reference standard	Adjudicated by ≥2 graders?	Clinical pathway	Image quality assessment	Time to result
Vaghefi E, 2021	New Zealand	Canon CR-2 AF, CR-2, Dgi, iCare EIDON TRC-NW65; Topcon, Tokyo, Japan	N/A	Mydriated if indicated by small pupil size	CNN, not specified	Not specified	Not specified	No	NZMH (38)	Grading by three senior retinal specialists	Yes	Hospital or optometrist clinic setting	Automated	Not specified
Wang Y, 2021	China	TRC-NW65; Topcon, Tokyo, Japan	N/A	Nonmydriatic	InceptionV3	Not specified	Not specified	No	NHS	Grading by three trained graders and two licensed ophthalmologists	Yes	Primary care centers	Automated	Time to grade image: 0.058 min for fully automated approach
Wongchaisuwat N, 2021	Thailand	Nonmydriatic AFC-330; NIDEK Co., Ltd	45°	Nonmydriatic	SE-ResNeX150_32 × 4d, SE-ResNeX101_32 × 4d, and SENet154	80% model training, 10% model validation, 10% model testing	Yes	No	Modified version of ETDRS	Retinal examination by retinal expert	Not specified	Hospital outpatient eye clinic and diabetes center	Manual	Not specified
Yang Y, 2022	China	Zeiss Visucam FF450 (mydriatic), Topcon TRC-50DX (mydriatic), Topcon NW-400	N/A	Mydriatic at two study centers and nonmydriatic at one study center	CNN, not specified	73,849 images in the training set and 4,928 images in the test set	Not specified	No	COS	Grading by two trained graders	Yes	Hospital clinic	Manual	Average diagnostic time of: AI – 24 s; gold standard grading – 69 s
Zhang Y, 2020	China	Topcon TRC-NW400, Miis DSC-200, Canon CR-2 PLUS AF, Canon CR-2 AF and Zeiss VISUCAM200	45°	Nonmydriatic	VoxelCloud Retina assembled from 6 neural networks; architecture: Inception-ResNet v2	Not specified	No	No	Self-modified version of ICDR	Grading by two specialist graders	Yes	Diabetes care centers (nonophthalmic specialized clinical setting)	Automated	Not specified
Zhu A, 2023	U.S.	Canon CR-2 Plus AF	45°	Nonmydriatic	ResNet v2 SELENA+	Not specified	Not specified	No	ICDR	Grading by one tele-ophthalmologist and one clinical ophthalmologist	Not specified	Community hospital	N/A	Grading times: 129 s (tele-ophthalmology grader), 68 s for clinical ophthalmologist, at 35.6 s for deep learning system
VTDR (n = 2 studies)														
Brant A, 2025	India	TRC-NW400 (Topcon Healthcare), NFC-700 (CrystalVue), and 3netra Classic HD (Forus Health)	45°	Nonmydriatic unless poor image quality necessitated mydriasis	Inception-V3	Development set divided into 80% training, 20% tuning, with separate external validation	Not specified	No	ICDR	Grading by three board certified ophthalmologists	Not specified	Vision centers, diabetes clinics, or tertiary hospital settings	N/A	Not specified
Ruamviboonsuk P, 2022	Thailand	Topcon Maestro 3D OCT-1, Topcon TRC-NW300, Nidek AFC-230, Nidek AFC-210, and Nidek AFC-300	45°	Nonmydriatic	CNN, not specified	Not specified	Not specified	No	ICDR	Grading by three board-certified retina specialists	Yes	Primary care sites	Automated	Not specified

Continued on p. 518

Continued on p. 518

Table 1—Continued

Table 1—Continued															
Basic information			Image acquisition			Model development and capabilities				Comparator		Screening setting and workflow			
Reference	Country	Imaging device(s)	Imaging field of view	Mydriasis, yes or no	Architecture	Cross-validation/ splitting	Fine tuning	Screening for multiple diseases	Diagnostic criteria	Reference standard	Adjudicated by ≥2 graders?	Clinical pathway	Image quality assessment	Time to result	
Any DR and RDR (n = 4 studies) Karabeg M, 2025	Norway	CLARUS TM 700, Zeiss , and iCare DRS Plus TrueColor 45° × 40° for iCare DRS Plus	133° cropped to 45° for CLARUS, and 45° × 40° for iCare DRS Plus	Nonmydriatic	CNN, not specified	Not specified	Not specified	Yes	ICDR	A consensus of health care professionals (optometrist, ophthalmic nurses (certified graders), and an experienced ophthalmologist)	Not specified	Hospital clinics and external imaging station	Manual	Not specified	
Ming S, 2021	China	Canon CR-2	45°	Nonmydriatic	CNN, not specified	Not specified	Not specified	No	ICDR	Grading by two licensed ophthalmologists	Yes	Primary care clinic	Manual	Not specified	
Quinn N, 2021	Australia	Canon CR-2	N/A	Mixed mydriatic and nonmydriatic	CNN, not specified	80% for training and testing; 20% for internal validation; separate external validation	Not specified	No	NHS	Grading by one trained grader	Not specified	Primary care	N/A	Not specified	
Wintergerst M, 2021	Germany	CenterVue	N/A	Nonmydriatic	EyeArt 2.1	Not specified	Not specified	No	ICDR	Grading by a trained grader and an ophthalmologist	Not specified	Primary care	Manual	Mean image acquisition time: 7.5 min	
RDR and VTDR (n = 2 studies) Ipp E, 2021	U.S.	Canon CR-2 AF or Canon CR-2 Plus AF; Canon USA Inc	45°	Nonmydriatic	EyeArt 2.1.0	Not specified	Not specified	No	ETDRS	Grading by two certified graders	Yes	Primary care, general ophthalmology, and retina specialty centers	N/A	Not specified	
Shah A, 2021	Spain	Topcon TRC-NW200	45°	Mydriatic	IDx-DR vX2.1	Not specified	Not specified	No	ICDR	Grading by three ophthalmologists	Yes	Primary care	Mixed	Device result output time: 1 min	
Any DR, RDR, and VTDR (n = 1 study) Zhang WF, 2022	China	Topcon TRC-NW6S, Canon CR2, and KOWA Nonmyd α-DIII 8300	40°–45°	Nonmydriatic	EyeWisdom with ResNet-34 for the quality control module, and Inception-V3 network for DR classification model	15,611 images in training set; 1,640 images in validation set; 3,252 images in internal test set	Not specified	No	ICDR	Grading by three senior physicians	Yes	Hospital	Automated	Not specified	
Any DR, RDR, VTDR, and diabetic maculopathy (n = 1 study) Dong X, 2022	China	RetiCam 3100	50°	Nonmydriatic	Inception-ResNetV2 on TensorFlow	10:1 split of training; tuning data sets	Not specified	Yes	ICDR	Grading by two ophthalmologists	Not specified	Community health care centers	Manual	Not specified	
Any DR, RDR, and diabetic maculopathy (n = 1 study) Tan-Torres III A, 2025	India	Topcon Maestro	N/A	Mydriatic	Inception-v4	Development set divided into 80% training; 20% tuning	Not specified	No	ICDR	Grading by two ophthalmologists and one optometrist	Not specified	Hospital clinic	Automated	Not specified	
COS, Chinese Ophthalmic Society Diabetic Retinopathy Grading Standards (39); ETDRS, Early Treatment Diabetic Retinopathy Study; N/A, not available; ResNet, residual network; VGGNet, visual geometry group network; NHS, National Health Service Criteria for Diabetic Retinopathy (U.K.) (40); NZMH, New Zealand Ministry of Health (38).															

COS, Chinese Ophthalmic Society Diabetic Retinopathy Grading Standards (39); ETDRS, Early Treatment Diabetic Retinopathy Study; N/A, not available; ResNet, residual network; VGGNet, visual geometry group network; NHS, National Health Service Criteria for Diabetic Retinopathy (U.K.) (40); NZMH, New Zealand Ministry of Health (38).

Table 2—Pooled diagnostic performance of AI algorithms in the detection of DR, RDR, and VTDR

Outcomes	No. of studies	Pooled AUROC	Pooled sensitivity (95% CI), %	Pooled specificity (95% CI), %	DOR (95% CI)
DR: yes/no	12	0.965	91.3 (85.4–95.0)	93.8 (87.2–97.1)	164.19 (67.56–399.01)
RDR: yes/no	39	0.959	91.7 (89.5–93.4)	91.3 (88.6–93.5)	131.50 (92.49–186.97)
VTDR: yes/no	6	0.974	90.3 (69.3–97.5)	95.4 (92.4–97.2)	201.19 (63.01–642.36)

hospitals) and/or tertiary care (i.e., specialist eye hospitals) outperformed those conducted in primary care (AUROC 0.960 vs. 0.944, respectively). Notably, studies including at least some element of mydriasis significantly outperformed those without mydriasis (AUROC 0.976 vs. 0.958, respectively). Those using partial or fully automated image quality assessment outperformed those that used manual quality assessment across all diagnostic performance parameters (AUROC 0.969 vs. 0.947, respectively). Studies with a larger sample size (greater than the median of 848 participants) had higher pooled discriminative performance in all parameters compared with studies with smaller-than-median sample sizes. Studies using ICDR as the grading criteria outperformed those using non-ICDR criteria (AUROC 0.962 vs. 0.956, respectively). Studies with ophthalmologists as the reference standard outperformed those with trained graders as the reference standard (AUROC 0.966 vs. 0.951, respectively). Studies using multiple CNNs outperformed those using a single CNN (AUROC 0.966 vs. 0.955, respectively). All HSROC plots for subgroup analyses are presented in Supplementary Fig. 7.

Economic Evaluation of DL-DR

Table 4 summarizes the 15 economic evaluation studies. Eight studies were conducted in high-income countries: one each in the U.K., Singapore, Australia, Norway, and Japan, and three in the U.S. Six studies were conducted in upper-middle-income countries: one each in Brazil and Thailand, and four in China. One study was conducted in India, a lower-middle-income country. Of the included studies, 10 reported outcomes as cost per quality-adjusted life year (QALY) gained, while 5 only provided data on cost savings. The most common perspectives were health care system ($n = 5$) and societal ($n = 5$). Other studies adopted a range of perspectives, including those of the health care payer, health care provider,

taxpayer, and patient. Two studies explored more than one perspective.

High-Income Countries

Overall, DL-assisted DR screening approaches demonstrated significant potential to be cost-effective compared with the standard of care in high-income countries. An automated DR image assessment system-based screening showed a cost reduction of 12.8–21.0% and 23.3% in the U.K. and the U.S. (18), respectively. In Singapore, a fully autonomous DL-DR strategy yielded a cost saving of 14.3%, and a semiautomated screening strategy was the least expensive compared with human graders and the fully automated screening strategy. A Norwegian pilot study focused on minority women and found cost savings of up to 52% with DL screening compared with manual screening. In Australia, primary care-based DL screening was estimated to be highly cost-effective over 40 years. Evidence from Japan also strongly supported the implementation of DL-DR screening. Point-of-care AI was also found to be cost-effective, even over a short 1-year time horizon, in pediatric diabetes populations in the U.S. (19). Separately, another U.S. study found DL-DR was not cost-effective in small pediatric cohorts but was cost-effective in larger patient cohorts ($n > 400$ patients), due to enhanced ability to leverage economies of scale (20).

Upper-Middle-Income Countries

DL-assisted strategies were also potentially cost-effective in upper-middle-income countries. In China, automated DR detection demonstrated high cost-effectiveness compared with ophthalmologist screening in either rural (21–23) or urban areas (23). In contrast, other studies did not show DL-DR screening to be cost-effective compared with the status quo. One study undertaken in China showed that DL-DR may not be cost-effective in low-income settings, due to the low labor costs, unless compliance rates increase (24). Similarly,

compliance with treatment was the most influential factor in the degree of DL's cost-effectiveness in a study based on Thailand's national DR screening program, ranging from US\$20–\$50, depending on compliance. A study conducted in Brazil also showed that DL-DR screening was less cost-effective than the standard of care (i.e., opportunistic ophthalmology referral-based DR screening) over a lifetime.

Lower-Middle-Income Countries

One study conducted in India indicated that, when compared with no screening, telescreening was the most cost-effective scenario, with DL-DR screening being the most cost-effective strategy only when 79% of the population living with diabetes achieved glycemic control.

CONCLUSIONS

Generating high-quality, robust, and clinically relevant evidence to support the role of AI in health care remains a key goal for many countries and systems; DL-based DR screening using FPs is no exception (25). The present meta-analysis demonstrates that DL-based models can achieve a high level (>90%) of discriminative accuracy in detecting DR across prospective validation studies. Factors that may affect DL models' real-world performance include study region, clinical pathway, mydriasis, image quality control, sample size, grading criteria, reference standard, and model architecture. Our economic evaluation shows that DL-based DR detection using FPs is likely to be cost-effective, particularly in high-income countries. Overall, our study demonstrates that incorporating DL-based models to interpret FPs may provide opportunities to deliver accurate, sustainable, and cost-effective DR screening.

Early recognition of DR, particularly RDR and VTDR, remains a public health priority, and our results indicate the potential of DL-DR systems to address this need. However, in our subgroup analyses of RDR studies, high levels of inter-subgroup variation

Table 3—Subgroup analysis for identifying RDR

Study region	Subgroup variables	No. of studies	Pooled AUROC	P value	Pooled sensitivity (95% CI), %	Pooled specificity (95% CI), %	DOR (95% CI)
Study region	Low- and lower-to-middle-income economies	5	0.916	<0.001	92.4 (91.3–93.3)	94.6 (91.4–96.7)	252.27 (116.47–546.42)
	Upper-middle- and high-income economies	34	0.957		91.5 (89.0–93.4)	90.7 (87.4–93.2)	118.84 (81.22–173.88)
Clinical pathway	Primary care	10	0.944	<0.001	87.9 (83.2–91.5)	91.8 (87.2–94.8)	96.58 (42.98–217.04)
	Secondary and/or tertiary care	22	0.960		92.2 (89.6–94.3)	90.4 (86.5–93.2)	124.96 (83.79–186.36)
Mydriasis	No mydriasis	29	0.958	<0.001	91.0 (88.6–93.0)	91.9 (89.0–94.1)	128.83 (88.46–187.64)
	Partial or full mydriasis	7	0.976		95.7 (84.7–98.9)	92.1 (82.8–96.5)	291.85 (51.79–1644.77)
Image quality assessment	Automated or partially automated	19	0.969	<0.001	93.9 (90.8–96.0)	91.6 (87.4–94.5)	200.37 (122.91–326.65)
	Manual	14	0.947		89.6 (85.8–92.4)	91.3 (86.0–94.7)	93.21 (51.23–169.61)
Sample size	Large (sample size >848)*	19	0.964	<0.01	91.7 (89.2–93.7)	93.4 (90.5–95.4)	169.10 (115.77–246.99)
	Small (sample size ≤848)*	20	0.950		90.9 (86.7–93.9)	88.6 (83.3–92.4)	89.15 (49.63–160.14)
Grading criteria	ICDR	17	0.962	<0.01	90.5 (85.0–94.1)	91.8 (88.7–94.1)	121.36 (61.95–237.76)
	Non-ICDR	19	0.956		91.7 (89.5–93.5)	92.2 (88.3–94.8)	140.65 (93.51–211.55)
Reference standard	Ophthalmologists	24	0.966	<0.001	92.3 (89.2–94.5)	92.2 (89.0–94.5)	155.76 (98.08–247.38)
	Trained graders	10	0.951		91.6 (87.2–94.6)	87.8 (79.2–93.1)	85.22 (60.78–119.51)
Architecture	Multiple CNN	17	0.966	<0.001	92.9 (89.5–95.2)	91.7 (87.9–94.4)	171.85 (90.82–325.17)
	Single CNN	22	0.955		91.1 (88.0–93.5)	91.0 (86.6–94.1)	115.03 (73.82–179.26)

*For sample size, the cutoff point of 848 participants was derived from taking the median.

in discriminative performance were noted, reflecting several challenges. Studies conducted in low- and lower-middle-income countries and primary care settings demonstrated relatively lower discriminative performance; this may be attributed to the shared challenges faced by these implementation settings, such as a lack of operator expertise (26,27), infrastructural deficiencies (28), and data imbalance. To address these challenges, we found that two add-on features to the screening process—mydriasis and image quality control modules—may improve discriminative accuracy by reducing ungradable images; in this meta-analysis, mydriatic studies and those using automated image quality assessment demonstrated significantly better performance than their counterparts ($P < 0.001$). Smaller-than-median sample sizes were also associated with decreased accuracy, underscoring the need for multicenter collaborations and advanced AI methods to address sample size limitations (29). In addition, variation in grading criteria and grader expertise contributed to performance heterogeneity, highlighting the importance of standardized evaluation protocols. Finally, studies leveraging multiple CNNs demonstrated superior performance than those using a single CNN, emphasizing the need for ensemble strategies to improve accuracy.

This meta-analysis also identified several deficiencies across current prospective validation studies, highlighting opportunities for future enhancement. A key gap lies in the incomplete reporting of diagnostic performance metrics across studies. For example, a few studies included in the meta-analysis reported the area under the precision-recall curve (AUPRC) alongside precision—metrics that can be influenced by DR prevalence and reflect the reliability of positive predictions. For a more comprehensive understanding of model performance, future studies should consistently report key diagnostic metrics, including AUROC, AUPRC, sensitivity, specificity, positive predictive value, negative predictive value, precision, and accuracy at the optimal threshold as defined by the Youden Index (30). For comparing different models, we suggest fixing sensitivity or specificity at the levels mandated by local DR screening guidelines and then benchmarking other metrics at this threshold to assess relative superiority. In addition, although we noted statistically significant differences

Table 4—Characteristics and the main conclusions of included studies on economic evaluation of DR detection

Reference	Country	Health evaluation plan	Target population	Modeling approach	Time horizon	Discount rate, cost/benefit, %	Perspective	Intervention	Comparators	Screening outcome	Currency, price date, and conversion	Study conclusions	Sensitivity analysis
High-income countries													
Tufail A, 2017	U.K.	Cost-effectiveness analysis	20,258 patients with diabetes mellitus	Decision-tree model	June 2013 to November 2014	3.50	N/A	ARIAS	Strategy 1: replacing initial manual grading with an ARIAS Strategy 2: using an ARIAS prior to manual grading Strategy 3: fully automated screening performed by an eye care professional	Presence of potential sight-threatening retinopathy or technical failure Number of true DR cases	US\$; 2014; UK pounds to US\$	ARIAS have the potential to reduce costs in developed-world health care economies	One-way sensitivity analysis
Wolf RM, 2020 (9)	U.S.	Cost-effectiveness analysis	Hypothetical cohort of youth (age <21 years) with T1D or T2D	Decision-tree model	March 2019 to January 2020	N/A	Patients' perspective	DL	Strategy 1: autonomous DL retinopathy screening Strategy 2: clinician-based screening examination performed by an eye care professional	Number of true DR cases	US\$; 2019; N/A	Point-of-care DR screening using autonomous DL systems is effective and cost saving for children with diabetes and their caregivers at recommended adherence rates.	One-way sensitivity analysis
Xie YC, 2020	Singapore	Cost-minimization analysis	39,006 consecutive patients with diabetes	Decision-tree model	12 months	N/A	The health system perspective	DL	Strategy 1: human graders Strategy 2: semiautomated screening Strategy 3: fully automated screening model	Number of patients referred and treated for DR	US\$; 2015; Singapore dollar \$1 per US\$0.75	From the health system perspective, the semiautomated screening model was the least expensive of the three models	Deterministic (one-way) sensitivity analyses
Fuller SD, 2022 (18)	U.S.	Cost-utility analysis	All adult patients with diabetes; primary care clinic	Markov model with microsimulation	5 years	3	N/A	ARIAS	Strategy 1: ARIAS Strategy 2: the standard of care, referring all patients with diabetes for a dilated screening eye examination	QALYs	US\$; 2019; N/A	Primary care-based ARIAS screening is cost-effective	One-way sensitivity analysis
Hu W, 2025 (35)	Australia	Cost-effectiveness analysis	1,197,818 non-Indigenous and 65,160 Indigenous Australians living with diabetes aged ≥20 years	Decision-analytic Markov model	40 years	3.5	Health care provider's perspective	DL	Scenario A: replacing clinician screening with DL in subpopulation currently under screening Scenario B: increasing screening rate to the patient acceptance level Scenario C: universal screening	QALYs	Australian dollars; N/A; N/A	The DL-powered intervention in all three scenarios was effective and cost-saving	One-way, two-way, and probabilistic sensitivity analysis
Karabeg M, 2024	Norway	Cost-minimization analysis	33 patients aged >18 years with diabetes mellitus	N/A	1 November 2017	N/A	Extended health care perspective	DL	Strategy 1: DL screening Strategy 2: manual human screening	Cost savings per patient	US\$; 2022; 9.89 Norwegian kroner per US\$1	The cost difference for DL screening between human and DL screening was \$143/patient, in favor of DL being more cost-saving	One-way deterministic analysis and two-way sensitivity analysis
Ahmed M, 2025 (20)	U.S.	Cost-effectiveness analysis	Hypothetical cohort of patients aged <21 years with T1D and T2D	Decision-analytic model	1 year	N/A	Health care system's perspective	DL	Strategy 1: Autonomous DL Strategy 2: standard of care referral to optometrist or ophthalmologist	ICER (cost savings)	US\$; N/A; N/A	In a single-practice cohort of 100 patients, DL use resulted in increased incremental costs compared with standard of care; in larger cohorts with >400 patients, DL use resulted in cost savings compared with standard of care	One-way sensitivity analysis
Akune Y, 2025	Japan	Cost-effectiveness analysis	Hypothetical cohort of 500,000 Japanese patients with T2D, aged 40 years	Individual-based state transition microsimulation model	From age 40 years until age 90 years, or until death	2	Health care payer's perspective	DL	Strategy 1: DL screening Strategy 2: conventional opportunistic DR screening	ICER (cost per QALY)	US\$; 2023; 140,511 Japanese yen per US\$1	DL screening was cost-effective compared with conventional screening.	One-way, two-way, and probabilistic sensitivity analysis
Upper-middle-income countries													
Gomez Rossi J, 2022 (36)	Brazil	Cost-effectiveness analysis	Individuals aged 40 years with diabetes	Markov model with microsimulation	Lifetime	3	Brazilian taxpayer's perspective	DL	Strategy 1: DL-supported decision-making Strategy 2: the standard screening of DR undertaken by ophthalmologists	QALYs	US\$; 2020; Brazilian Real \$2.362 per US\$1	The DL solutions used as decision-support systems showed only moderate cost-effectiveness improvement	One-way sensitivity analysis

Continued on p. 522

Table 4—Continued

Reference	Country	Health evaluation plan	Target population	Modeling approach	Time horizon	Discount rate, cost/benefit, %	Perspective	Intervention	Comparators	Screening outcome	Currency, price date, and conversion	Study conclusions	Sensitivity analysis
Huang XM, 2022 (21)	China	Cost-effectiveness analysis	Hypothetical cohort of 1,000 patients with newly diagnosed diabetes but without DR	Markov model-based hybrid decision tree	35 years	3	The health system perspective and the societal perspective	DL	Strategy 1: DL screening Strategy 2: no screening Strategy 3: ophthalmologists screening strategies	QALYs	US\$; 2020; Chinese yuan 6.9129 per US\$1	DL-based screening is more cost-effective	One-way sensitivity analysis and probabilistic analysis
Li H, 2023 (22)	China	Cost-effectiveness analysis	10,000 patients with diabetes mellitus in rural China	Markov model	50 years	3	Health system perspective	DL	Strategy 1: DL screening Strategy 2: no screening Strategy 3: ophthalmologist screening	QALYs	US\$; N/A; 7.189 Chinese yuan per US\$1	Compared with ophthalmologist screening and no screening for DR, DL-based screening was the most effective, including cost-savings and improvement to quality of life in rural areas	One-way sensitivity analysis and probabilistic sensitivity analysis
Lin SL, 2023 (24)	China	Cost-effectiveness and cost-utility analysis	Hypothetical cohort of community residents aged ≥65 years, with diabetes mellitus, in an LMIC setting	Decision-analytic Markov models	30 years	3.5	Societal perspective	DL	Strategy 1: manual grading by trained grader (telemedicine-based) Strategy 2: DL-assisted DR telemedicine screening	QALYs	US\$; Chinese yuan 6.9 per US\$1	DL-based screening may not prove cost-effective in LMIC settings where labor costs are low, unless compliance with referrals improves	One-way sensitivity analysis
Srisubut A, 2023	Thailand	Cost-utility analysis	Hypothetical cohort of patients with T2D, aged 40 years, not previously screened for DR	Markov model with embedded decision tree	Lifetime	3	Societal and provider perspectives	DL	Strategy 1: trained manual graders Strategy 2: grading using DL	QALYs	US\$; 32.02 Thai baht per US\$1	DL-based DR screening in a MIC may lower societal costs over a lifetime. Compared with trained graders, DL may cause a higher ICER from a provider perspective	One-way sensitivity analysis
Wang Y, 2024 (23)	China	Cost-effectiveness analysis	251,535 participants with diabetes mellitus	Hybrid decision tree/Markov model	30 years	3	Societal perspective	DL	1,100 different diagnostic performances (presented as sensitivity/specificity pairs) and modeled annual screening scenarios	QALYs	US\$; 2019; 6.8968 Chinese yuan per US\$1	DL models should have a minimum sensitivity of 88.2% and specificity of 80.4% for cost-savings and to be cost-effective. In settings with higher DR prevalence and willingness to pay, urban regions, and younger patients, the DL needed higher sensitivity for optimal cost-effectiveness	Univariate sensitivity analysis and probabilistic sensitivity analysis
Lower-middle-income countries													
Purohit N, 2025	India	Cost-effectiveness analysis	Hypothetical cohort of 1,000 patients aged 45 years, with diabetes	Markov model and decision tree	Lifetime	3	Societal perspective	DL	Scenario 1: face-to-face screening Scenario 2: teleophthalmology-supported screening Scenario 3: DL-supported screening Scenario 4: usual-care scenario (i.e., no population screening)	ICER (cost per QALY gained)	US\$; 78.60 Indian rupees per US\$1	Annual tele-supported (no DL) screening for individuals living with diabetes for ≥5 years is the most cost-effective strategy; if adequate glycemic control was achieved in 79% of the population, annual DL-supported screening of individuals living with diabetes for >10 years becomes the most effective strategy	Multivariate and univariate probabilistic sensitivity analysis

ARIAS, automated retinal image analysis system; ICER, incremental cost-effectiveness ratio; LMIC, low-to-middle-income country; MIC, middle-income country; N/A, not available; T1D, type 1 diabetes; T2D, type 2 diabetes.

between subgroups in our meta-analysis, this does not always translate to clinical significance. To that end, we also recommend reporting other complementary performance metrics, such as decision curve analysis and net benefit, because these metrics are increasingly recognized as more appropriate for capturing the practical utility of models in risk-based screening and triage settings (31,32). Another gap relates to patient characterization: >75% of studies did not report the burden of ocular or systemic comorbid conditions, limiting contextual interpretation of diagnostic performance (Supplementary Table 5). Standardized reporting of patient characteristics would improve comparability and external validity. An additional gap concerns model development practices, because most studies used simple train-test splits instead of cross-validation. Work comparing the robustness and reproducibility of cross-validated algorithms against those using train-test splits could help identify whether cross-validation confers a real-world advantage and guide training strategy selection for DL-DR systems. Reporting practices for model development were also inconsistent, with few studies specifying whether models were locked before prospective validation or were fine tuned. Greater transparency in this regard would enable clinicians to better assess model generalizability. Moreover, in settings where large-scale local data acquisition is challenging, evaluating the utility of limited local data for fine-tuning or domain adaptation strategies (e.g., test-time adaptation [33]) may be critical for optimizing performance across diverse populations, imaging devices, and screening environments.

Aside from performance, successful deployment of DL-based DR detection also depends on strategic alignment across the “5Ps”: people, policies, processes, platforms, and products (34). Applying this lens, we also provide recommendations to fill current gaps for future implementation. First, our evidence was mostly derived from high-income countries or those with a well-established screening infrastructure, underscoring the need for studies conducted in low- and lower-middle-income countries to enhance model generalizability and facilitate equitable access to DL-based DR screening worldwide. Second, current DL models remain imperfect due to misclassifications arising from key

contributors, including the presence of peripheral lesions, overlying epiretinal membranes, confusion between minor hemorrhages and microaneurysms, and confusion between DME and drusen. Clinicians should be aware of these limitations and ensure cross-checking when interpreting DL outputs. Deployment strategies should also incorporate follow-up pathways or safety nets (33) to mitigate the clinical consequences of misclassifications. Third, only four studies included in our review evaluated end-users’ feedback during the prospective validation. Data collection encompassing dimensions such as usability, feasibility, and end-user acceptance is critical for meaningful deployment of DL-based DR detection (35). Fourth, more than half of the studies did not report explainability methods, such as heat maps or lesion atlases; incorporating explainability measures could enhance transparency and foster greater trust and acceptance among clinicians, patients, and policymakers.

It is crucial to emphasize that a DL model’s high diagnostic performance does not necessarily translate to high economic value—an important consideration for health policy implementation. To that end, we also provide an updated synthesis of economic evaluation studies. Overall, we found that DL-based DR detection was cost-saving in comparison with conventional manual DR detection. However, to augment cost-effectiveness, especially for low- and lower-middle-income countries, it is essential to improve patients’ compliance and optimize glycemic control, as well as reduce initial implementation costs. Notably, there are still gaps to fill based on current economic analysis studies. First, not all studies conducted cost-effectiveness analyses using standardized outcomes (e.g., cost-per-QALY), with some merely reporting cost comparisons. For enhanced evidence synthesis, studies should aim to report cost-per-QALY. Second, a major limitation of economic analyses conducted in upper-middle-income countries is the use of data based on developed-world economies, which do not account for regional differences in resource costs (36), disease prevalence, and willingness to pay. To provide a more representative evaluation of AI-based DR screening in upper-middle-income countries, studies should aim to use local data as input parameters, which

will also aid the selection of an optimal DL model to fit the local population. Third, there remains a lack of evidence on cost-effectiveness analysis in low-income countries. Investigations covering more income levels and different settings are still warranted. Fourth, there was marked variation in the perspectives adopted, bearing important implications for interpreting cost-effectiveness. For example, DL-DR screening that appears favorable from a societal perspective, capturing productivity gains and informal care, may not be cost-effective from a narrower health care payer perspective. Where possible, reporting multiple perspectives could enhance the comparability and generalizability of findings for policy decisions.

Overall, our study has several strengths. First, we highlighted DL models’ real-world utility for DR screening in clinical practice by synthesizing prospective validation studies comprising 329,487 images, eyes, and participants. Second, we identified significant factors affecting model performance and existing gaps, as well as provided recommendations to address these factors prior to the real-world deployment of DL-enabled pathways. Third, we included 15 studies for a structured, up-to-date synthesis of economic analyses that capture key trends in the cost-effectiveness of DR deployment across countries, contrast and compare methodologies used, and outline gaps in the design and data collection among economic evaluation studies. Fourth, our study includes data spanning high- and low- to middle-income settings, covering 20 countries across 6 continents. We recognized unmet needs for DL model development in low-income countries, enabling us to identify targeted areas for future work. More importantly, the integrated review of diagnostic performance and cost-effectiveness can potentially offer clinicians and policymakers actionable insights for implementation.

Our study also has limitations. First, a paucity of studies assessing multiple severities of DR, especially VTDR and any DR, restricted our ability to perform comparative analyses in such cohorts. Given the clinical importance of VTDR, prospective studies should aim to separately report AI performance metrics for varying DR severities. Second, we could not perform a separate subgroup analysis on DME, due to the lack of data within the studies included in our meta-analysis

(37). Since DME is of high clinical importance due to its threat to central vision, it would be more useful for prospective validation studies to report DME-specific data. Third, insights into low- and lower-middle-income countries were limited by a lack of studies in these regions, indicating a need for further data acquisition. Fourth, this study only included publications written in English. Given the global burden of DR, future research should aim to incorporate studies published in a wider range of languages.

Our systematic review and meta-analysis demonstrate that DL-DR systems using FPs had high overall discriminative performance in prospective real-world settings. DL-DR systems also hold promise to improve the cost-effectiveness of DR screening programs, especially in high-income countries.

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References

- Sivaprasad S, Wong TY, Gardner TW, Sun JK, Bressler NM. Diabetic retinal disease. *Nat Rev Dis Primers* 2025;11:62
- Burton MJ, Ramke J, Marques AP, et al. The Lancet Global Health Commission on Global Eye Health: vision beyond 2020. *Lancet Glob Health* 2021;9:e489–e551
- Wang LZ, Cheung CY, Tapp RJ, et al. Availability and variability in guidelines on diabetic retinopathy screening in Asian countries. *Br J Ophthalmol* 2017;101:1352–1360
- Wong TY, Sun J, Kawasaki R, et al. Guidelines on diabetic eye care: the International Council of Ophthalmology recommendations for screening, follow-up, referral, and treatment based on resource settings. *Ophthalmology* 2018;125:1608–1622
- Scanlon PH, Aldington SJ, Leal J, et al. Development of a cost-effectiveness model for optimisation of the screening interval in diabetic retinopathy screening. *Health Technol Assess* 2015;19:1–116
- Ting DSW, Cheung CY-L, Lim G, et al. Development and validation of a deep learning system for diabetic retinopathy and related eye diseases using retinal images from multiethnic populations with diabetes. *JAMA* 2017;318:2211–2223
- Verbraak FD, Abramoff MD, Bausch GCF, et al. Diagnostic accuracy of a device for the automated detection of diabetic retinopathy in a primary care setting. *Diabetes Care* 2019;42:651–656
- Lee AY, Yanagihara RT, Lee CS, et al. Multicenter, head-to-head, real-world validation study of seven automated artificial intelligence diabetic retinopathy screening systems. *Diabetes Care* 2021;44:1168–1175
- Rajesh AE, Davidson OQ, Lee CS, Lee AY. Artificial intelligence and diabetic retinopathy: AI framework, prospective studies, head-to-head validation, and cost-effectiveness. *Diabetes Care* 2023;46:1728–1739
- Shamseer L, Moher D, Clarke M, et al.; PRISMA-P Group. Preferred Reporting Items for Systematic Review and Meta-analysis Protocols (PRISMA-P) 2015: elaboration and explanation. *BMJ* 2015;350:g7647
- Wilkinson CP, Ferris FL, Klein RE, et al.; Global Diabetic Retinopathy Project Group. Proposed international clinical diabetic retinopathy and diabetic macular edema disease severity scales. *Ophthalmology* 2003;110:1677–1682
- The World Bank. The World Bank Atlas method - detailed methodology. Accessed 18 August 2025. Available from <https://datahelpdesk.worldbank.org/knowledgebase/articles/378832-what-is-the-world-bank-atlas-method>
- Glas AS, Lijmer JG, Prins MH, Bossuyt PMM. The diagnostic odds ratio: a single indicator of test performance. *J Clin Epidemiol* 2003;56:1129–1135
- Deeks JJ, Macaskill P, Irwig L. The performance of tests of publication bias and other sample size effects in systematic reviews of diagnostic test accuracy was assessed. *J Clin Epidemiol* 2005;58:882–893
- Lowry R. VassarStats: website for statistical computation, 2023. Accessed 25 August 2025. Available from vassarstats.net/
- Elvidge J, Hawksworth C, Avşar TS, et al.; CHEERS-AI Steering Group. Consolidated Health Economic Evaluation Reporting Standards for Interventions That Use Artificial Intelligence (CHEERS-AI). *Value Health* 2024;27:1196–1205
- Lin L, Chu H, Murad MH, et al. Empirical comparison of publication bias tests in meta-analysis. *J Gen Intern Med* 2018;33:1260–1267
- Fuller SD, Hu J, Liu JC, et al. Five-year cost-effectiveness modeling of primary care-based, nonmydriatic automated retinal image analysis screening among low-income patients with diabetes. *J Diabetes Sci Technol* 2022;16:415–427
- Wolf RM, Channa R, Abramoff MD, Lehmann HP. Cost-effectiveness of autonomous point-of-care diabetic retinopathy screening for pediatric patients with diabetes. *JAMA Ophthalmol* 2020;138:1063–1069
- Ahmed M, Dai T, Channa R, Abramoff MD, Lehmann HP, Wolf RM. Cost-effectiveness of AI for pediatric diabetic eye exams from a health system perspective. *NPJ Digit Med* 2025;8:3
- Huang X-M, Yang B-F, Zheng W-L, et al. Cost-effectiveness of artificial intelligence screening for diabetic retinopathy in rural China. *BMC Health Serv Res* 2022;22:260
- Li H, Li G, Li N, et al. Cost-effectiveness analysis of artificial intelligence-based diabetic retinopathy screening in rural China based on the Markov model. *PLoS One* 2023;18:e0291390
- Wang Y, Liu C, Hu W, et al. Economic evaluation for medical artificial intelligence: accuracy vs. cost-effectiveness in a diabetic retinopathy screening case. *NPJ Digit Med* 2024;7:43
- Lin S, Ma Y, Xu Y, et al. Artificial intelligence in community-based diabetic retinopathy telemedicine screening in urban China: cost-effectiveness and cost-utility analyses with real-world data. *JMIR Public Health Surveill* 2023;9:e41624
- Macdonald T, Zhelev Z, Liu X, et al. Generating evidence to support the role of AI in diabetic eye screening: considerations from the UK National Screening Committee. *Lancet Digit Health* 2025;7:100840
- Gu C, Wang Y, Jiang Y, et al. Application of artificial intelligence system for screening multiple fundus diseases in Chinese primary healthcare settings: a real-world, multicentre and cross-sectional study of 4795 cases. *Br J Ophthalmol* 2024;108:424–431
- Dong X, Du S, Zheng W, Cai C, Liu H, Zou J. Evaluation of an artificial intelligence system for the detection of diabetic retinopathy in Chinese community healthcare centers. *Front Med (Lausanne)* 2022;9:883462
- Ruamviboonsuk P, Tiwari R, Sayres R, et al. Real-time diabetic retinopathy screening by deep learning in a multisite national screening programme: a prospective interventional cohort study. *Lancet Digit Health* 2022;4
- Tham YC, Goh JHL, Nderitu P, et al.; Global RETFound Consortium. Building the world's first truly global medical foundation model. *Nat Med* 2025;31:3580–3585
- Ruopp MD, Perkins NJ, Whitcomb BW, Schisterman EF. Youden Index and optimal cut-point estimated from observations affected by a lower limit of detection. *Biom J* 2008;50:419–430
- Zhang J, Lin S, Cheng T, et al. RETFound-enhanced community-based fundus disease screening: real-world evidence and decision curve analysis. *NPJ Digit Med* 2024;7:108
- Sounderajah V, Guni A, Liu X, et al.; STARD-AI Steering Committee. The STARD-AI reporting guideline for diagnostic accuracy studies using artificial intelligence. *Nat Med* 2025;10.1038/s41591
- Nguyen TX, Jiang M, Yang D, et al. Advancing diabetic macular edema detection from 3d optical coherence tomography scans: integrating privacy-preserving AI and generalizability techniques — a prospective validation in Vietnam. *NEJM AI* 2024;1A042400091
- Gunasekaran DV, Miller S, Hsu W, et al. National use of artificial intelligence for eye screening in Singapore. *Nejm AI* 2024;1A1cs2400404

35. Hu W, Lin Z, Clark M, et al. Real-world feasibility, accuracy and acceptability of automated retinal photography and AI-based cardiovascular disease risk assessment in Australian primary care settings: a pragmatic trial. *NPJ Digit Med* 2025;8:122
36. Gomez Rossi J, Rojas-Perilla N, Krois J, Schwendicke F. Cost-effectiveness of artificial intelligence as a decision-support system applied to the detection and grading of melanoma, dental caries, and diabetic retinopathy. *JAMA Netw Open* 2022;5:e220269
37. Chawla S, Chawla A, Chawla R, Jaggi S, Singh D, Trehan S. Trained nurse-operated teleophthalmology screening approach as a cost-effective tool for diabetic retinopathy. *Int J Diabetes Dev Ctries* 2022;42:747–750
38. Ministry of Health. Diabetic Retinal Screening, Grading, Monitoring and Referral Guidance, 2016. Accessed 15 August 2025. Available from <https://www.tewhatuora.govt.nz/assets/Publications/Diabetes/diabetic-retinal-screening-grading-monitoring-referral-guidance-mar16.pdf>
39. Yang Y, Pan J, Yuan M, et al. Performance of the AIDRScreening system in detecting diabetic retinopathy in the fundus photographs of Chinese patients: a prospective, multicenter, clinical study. *Ann Transl Med* 2022;10:1088
40. Scanlon PH. The English National Screening Programme for diabetic retinopathy 2003–2016. *Acta Diabetol* 2017;54:515–525